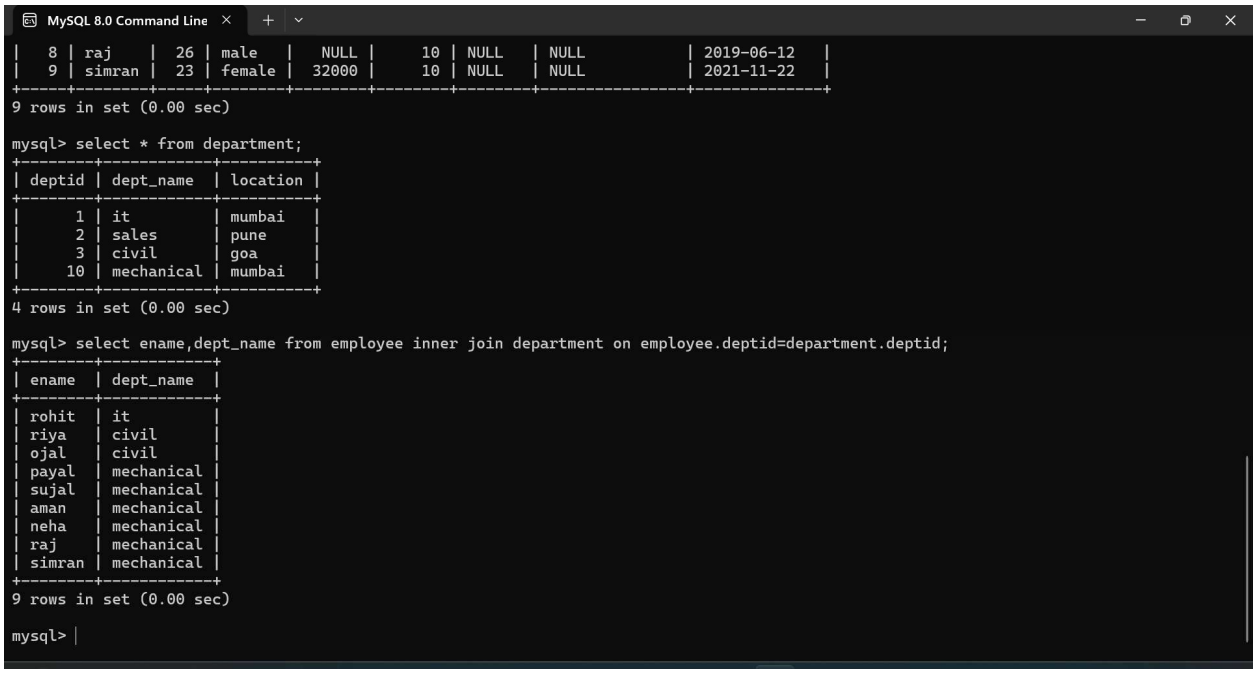
Assignment 4

Q1.

```
mysql> select * from employee cross join department;
```

eid	ename	age	gender	salary	deptid	city	DepartmentName	joining_date	deptid	dept_name	location
1	payal	25	female	27500	10	Mumbai	sales	2026-10-01	10	mechanical	mumbai
1	payal	25	female	27500	10	Mumbai	sales	2026-10-01	3	civil	goa
1	payal	25	female	27500	10	Mumbai	sales	2026-10-01	2	sales	pune
1	payal	25	female	27500	10	Mumbai	sales	2026-10-01	1	it	mumbai
2	sujal	23	male	22000	10	goa	NULL	2026-10-05	10	mechanical	mumbai
2	sujal	23	male	22000	10	goa	NULL	2026-10-05	3	civil	goa
2	sujal	23	male	22000	10	goa	NULL	2026-10-05	2	sales	pune
2	sujal	23	male	22000	10	goa	NULL	2026-10-05	1	it	mumbai
3	riya	28	female	50000	3	pune	NULL	2026-10-10	10	mechanical	mumbai
3	riya	28	female	50000	3	pune	NULL	2026-10-10	3	civil	goa
3	riya	28	female	50000	3	pune	NULL	2026-10-10	2	sales	pune
3	riya	28	female	50000	3	pune	NULL	2026-10-10	1	it	mumbai
4	rohit	28	male	NULL	1	mumbai	NULL	2026-09-10	10	mechanical	mumbai
4	rohit	28	male	NULL	1	mumbai	NULL	2026-09-10	3	civil	goa
4	rohit	28	male	NULL	1	mumbai	NULL	2026-09-10	2	sales	pune
4	rohit	28	male	NULL	1	mumbai	NULL	2026-09-10	1	it	mumbai
5	ojal	45	male	56000	3	pune	NULL	2026-09-23	10	mechanical	mumbai
5	ojal	45	male	56000	3	pune	NULL	2026-09-23	3	civil	goa
5	ojal	45	male	56000	3	pune	NULL	2026-09-23	2	sales	pune
5	ojal	45	male	56000	3	pune	NULL	2026-09-23	1	it	mumbai
6	aman	24	male	25000	10	NULL	NULL	2020-01-02	10	mechanical	mumbai
6	aman	24	male	25000	10	NULL	NULL	2020-01-02	3	civil	goa
6	aman	24	male	25000	10	NULL	NULL	2020-01-02	2	sales	pune
6	aman	24	male	25000	10	NULL	NULL	2020-01-02	1	it	mumbai
7	neha	25	female	30000	10	NULL	NULL	2020-06-12	10	mechanical	mumbai
7	neha	25	female	30000	10	NULL	NULL	2020-06-12	3	civil	goa
7	neha	25	female	30000	10	NULL	NULL	2020-06-12	2	sales	pune
7	neha	25	female	30000	10	NULL	NULL	2020-06-12	1	it	mumbai
8	raj	26	male	NULL	10	NULL	NULL	2019-06-12	10	mechanical	mumbai
8	raj	26	male	NULL	10	NULL	NULL	2019-06-12	3	civil	goa
8	raj	26	male	NULL	10	NULL	NULL	2019-06-12	2	sales	pune
8	raj	26	male	NULL	10	NULL	NULL	2019-06-12	1	it	mumbai
9	simran	23	female	32000	10	NULL	NULL	2021-11-22	10	mechanical	mumbai
9	simran	23	female	32000	10	NULL	NULL	2021-11-22	3	civil	goa
9	simran	23	female	32000	10	NULL	NULL	2021-11-22	2	sales	pune
9	simran	23	female	32000	10	NULL	NULL	2021-11-22	1	it	mumbai

Q2.



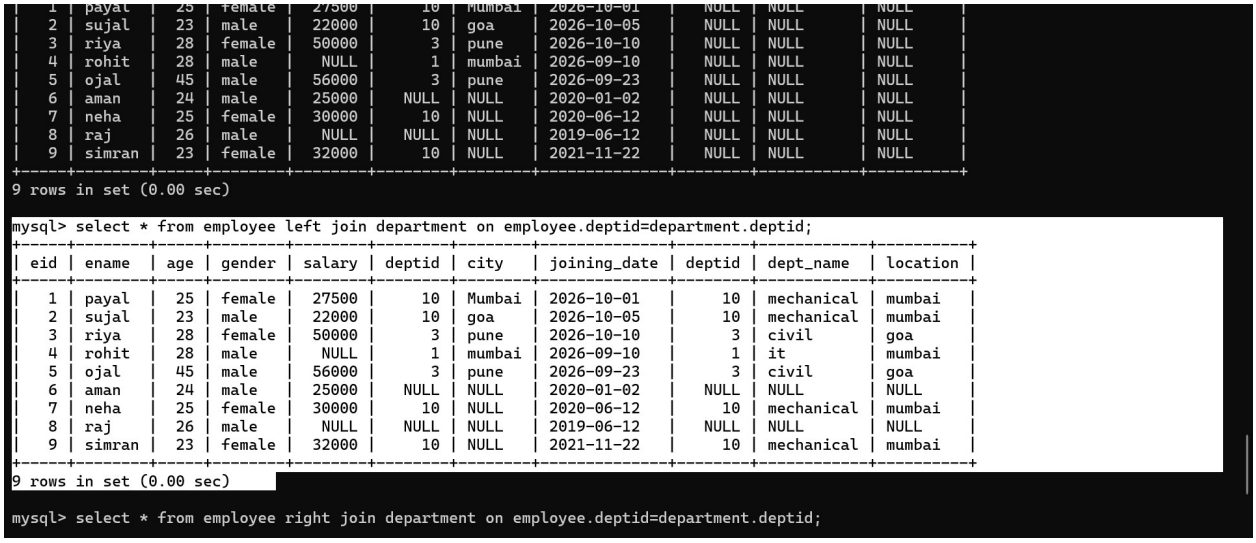
```
MySQL 8.0 Command Line
|   8 | raj    |  26 | male   |   NULL |     10 | NULL   | NULL           | 2019-06-12   |
|   9 | simran |  23 | female |  32000 |     10 | NULL   | NULL           | 2021-11-22   |
+-----+--------+-----+--------+--------+--------+--------+----------------+--------------+
9 rows in set (0.00 sec)

mysql> select * from department;
+--------+------------+----------+
| deptid | dept_name  | location |
+--------+------------+----------+
|      1 | it         | mumbai   |
|      2 | sales      | pune     |
|      3 | civil      | goa      |
|     10 | mechanical | mumbai   |
+--------+------------+----------+
4 rows in set (0.00 sec)

mysql> select ename,dept_name from employee inner join department on employee.deptid=department.deptid;
+--------+------------+
| ename  | dept_name  |
+--------+------------+
| rohit  | it         |
| riya   | civil      |
| ojal   | civil      |
| payal  | mechanical |
| sujal  | mechanical |
| aman   | mechanical |
| neha   | mechanical |
| raj    | mechanical |
| simran | mechanical |
+--------+------------+
9 rows in set (0.00 sec)

mysql>
```

Q3.



```
|   1 | payal  |  25 | female |  27500 |     10 | Mumbai | 2026-10-01   |   NULL | NULL       | NULL     |
|   2 | sujal  |  23 | male   |  22000 |     10 | goa    | 2026-10-05   |   NULL | NULL       | NULL     |
|   3 | riya   |  28 | female |  50000 |      3 | pune   | 2026-10-10   |   NULL | NULL       | NULL     |
|   4 | rohit  |  28 | male   |   NULL |      1 | mumbai | 2026-09-10   |   NULL | NULL       | NULL     |
|   5 | ojal   |  45 | male   |  56000 |      3 | pune   | 2026-09-23   |   NULL | NULL       | NULL     |
|   6 | aman   |  24 | male   |  25000 |   NULL | NULL   | 2020-01-02   |   NULL | NULL       | NULL     |
|   7 | neha   |  25 | female |  30000 |     10 | NULL   | 2020-06-12   |   NULL | NULL       | NULL     |
|   8 | raj    |  26 | male   |   NULL |   NULL | NULL   | 2019-06-12   |   NULL | NULL       | NULL     |
|   9 | simran |  23 | female |  32000 |     10 | NULL   | 2021-11-22   |   NULL | NULL       | NULL     |
+-----+--------+-----+--------+--------+--------+--------+--------------+--------+------------+----------+
9 rows in set (0.00 sec)

mysql> select * from employee left join department on employee.deptid=department.deptid;
+-----+--------+-----+--------+--------+--------+--------+--------------+--------+------------+----------+
| eid | ename  | age | gender | salary | deptid | city   | joining_date | deptid | dept_name  | location |
+-----+--------+-----+--------+--------+--------+--------+--------------+--------+------------+----------+
|   1 | payal  |  25 | female |  27500 |     10 | Mumbai | 2026-10-01   |     10 | mechanical | mumbai   |
|   2 | sujal  |  23 | male   |  22000 |     10 | goa    | 2026-10-05   |     10 | mechanical | mumbai   |
|   3 | riya   |  28 | female |  50000 |      3 | pune   | 2026-10-10   |      3 | civil      | goa      |
|   4 | rohit  |  28 | male   |   NULL |      1 | mumbai | 2026-09-10   |      1 | it         | mumbai   |
|   5 | ojal   |  45 | male   |  56000 |      3 | pune   | 2026-09-23   |      3 | civil      | goa      |
|   6 | aman   |  24 | male   |  25000 |   NULL | NULL   | 2020-01-02   |   NULL | NULL       | NULL     |
|   7 | neha   |  25 | female |  30000 |     10 | NULL   | 2020-06-12   |     10 | mechanical | mumbai   |
|   8 | raj    |  26 | male   |   NULL |   NULL | NULL   | 2019-06-12   |   NULL | NULL       | NULL     |
|   9 | simran |  23 | female |  32000 |     10 | NULL   | 2021-11-22   |     10 | mechanical | mumbai   |
+-----+--------+-----+--------+--------+--------+--------+--------------+--------+------------+----------+
9 rows in set (0.00 sec)

mysql> select * from employee right join department on employee.deptid=department.deptid;
```

Q4.

```
8 rows in set (0.00 sec)

mysql> select * from employee right join department on employee.deptid=department.dept.id;
ERROR 1054 (42S22): Unknown column 'department.dept.id' in 'on clause'
mysql> select * from employee right join department on employee.deptid=department.deptid;
+------+--------+------+--------+--------+--------+--------+--------------+--------+------------+----------+
| eid  | ename  | age  | gender | salary | deptid | city   | joining_date | deptid | dept_name  | location |
+------+--------+------+--------+--------+--------+--------+--------------+--------+------------+----------+
|    4 | rohit  |   28 | male   |   NULL |      1 | mumbai | 2026-09-10   |      1 | it         | mumbai   |
| NULL | NULL   | NULL | NULL   |   NULL |   NULL | NULL   | NULL         |      2 | sales      | pune     |
|    3 | riya   |   28 | female |  50000 |      3 | pune   | 2026-10-10   |      3 | civil      | goa      |
|    5 | ojal   |   45 | male   |  56000 |      3 | pune   | 2026-09-23   |      3 | civil      | goa      |
|    1 | payal  |   25 | female |  27500 |     10 | Mumbai | 2026-10-01   |     10 | mechanical | mumbai   |
|    2 | sujal  |   23 | male   |  22000 |     10 | goa    | 2026-10-05   |     10 | mechanical | mumbai   |
|    7 | neha   |   25 | female |  30000 |     10 | NULL   | 2020-06-12   |     10 | mechanical | mumbai   |
|    9 | simran |   23 | female |  32000 |     10 | NULL   | 2021-11-22   |     10 | mechanical | mumbai   |
+------+--------+------+--------+--------+--------+--------+--------------+--------+------------+----------+
8 rows in set (0.00 sec)

mysql>
```

Q5.

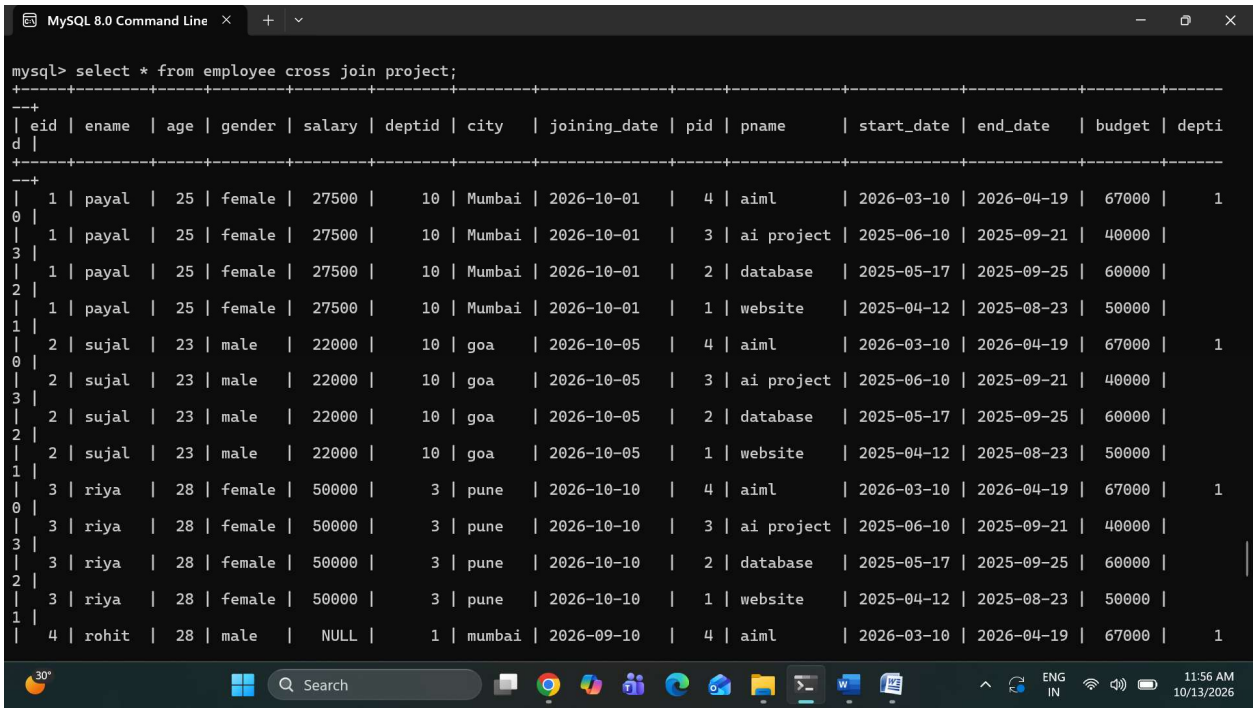
```
MySQL 8.0 Command Line

mysql> select * from employee right join department on employee.deptid=department.dept.id;
ERROR 1054 (42S22): Unknown column 'department.dept.id' in 'on clause'
mysql> select * from employee right join department on employee.deptid=department.deptid;
+------+--------+------+--------+--------+--------+--------+--------------+--------+------------+----------+
| eid  | ename  | age  | gender | salary | deptid | city   | joining_date | deptid | dept_name  | location |
+------+--------+------+--------+--------+--------+--------+--------------+--------+------------+----------+
|    4 | rohit  |   28 | male   |   NULL |      1 | mumbai | 2026-09-10   |      1 | it         | mumbai   |
| NULL | NULL   | NULL | NULL   |   NULL |   NULL | NULL   | NULL         |      2 | sales      | pune     |
|    3 | riya   |   28 | female |  50000 |      3 | pune   | 2026-10-10   |      3 | civil      | goa      |
|    5 | ojal   |   45 | male   |  56000 |      3 | pune   | 2026-09-23   |      3 | civil      | goa      |
|    1 | payal  |   25 | female |  27500 |     10 | Mumbai | 2026-10-01   |     10 | mechanical | mumbai   |
|    2 | sujal  |   23 | male   |  22000 |     10 | goa    | 2026-10-05   |     10 | mechanical | mumbai   |
|    7 | neha   |   25 | female |  30000 |     10 | NULL   | 2020-06-12   |     10 | mechanical | mumbai   |
|    9 | simran |   23 | female |  32000 |     10 | NULL   | 2021-11-22   |     10 | mechanical | mumbai   |
+------+--------+------+--------+--------+--------+--------+--------------+--------+------------+----------+
8 rows in set (0.00 sec)

mysql> select * from employee natural join department;
+--------+-----+--------+-----+--------+--------+--------+--------------+------------+----------+
| deptid | eid | ename  | age | gender | salary | city   | joining_date | dept_name  | location |
+--------+-----+--------+-----+--------+--------+--------+--------------+------------+----------+
|      1 |   4 | rohit  |  28 | male   |   NULL | mumbai | 2026-09-10   | it         | mumbai   |
|      3 |   3 | riya   |  28 | female |  50000 | pune   | 2026-10-10   | civil      | goa      |
|      3 |   5 | ojal   |  45 | male   |  56000 | pune   | 2026-09-23   | civil      | goa      |
|     10 |   1 | payal  |  25 | female |  27500 | Mumbai | 2026-10-01   | mechanical | mumbai   |
|     10 |   2 | sujal  |  23 | male   |  22000 | goa    | 2026-10-05   | mechanical | mumbai   |
|     10 |   7 | neha   |  25 | female |  30000 | NULL   | 2020-06-12   | mechanical | mumbai   |
|     10 |   9 | simran |  23 | female |  32000 | NULL   | 2021-11-22   | mechanical | mumbai   |
+--------+-----+--------+-----+--------+--------+--------+--------------+------------+----------+
7 rows in set (0.00 sec)

mysql>
```

Q6.

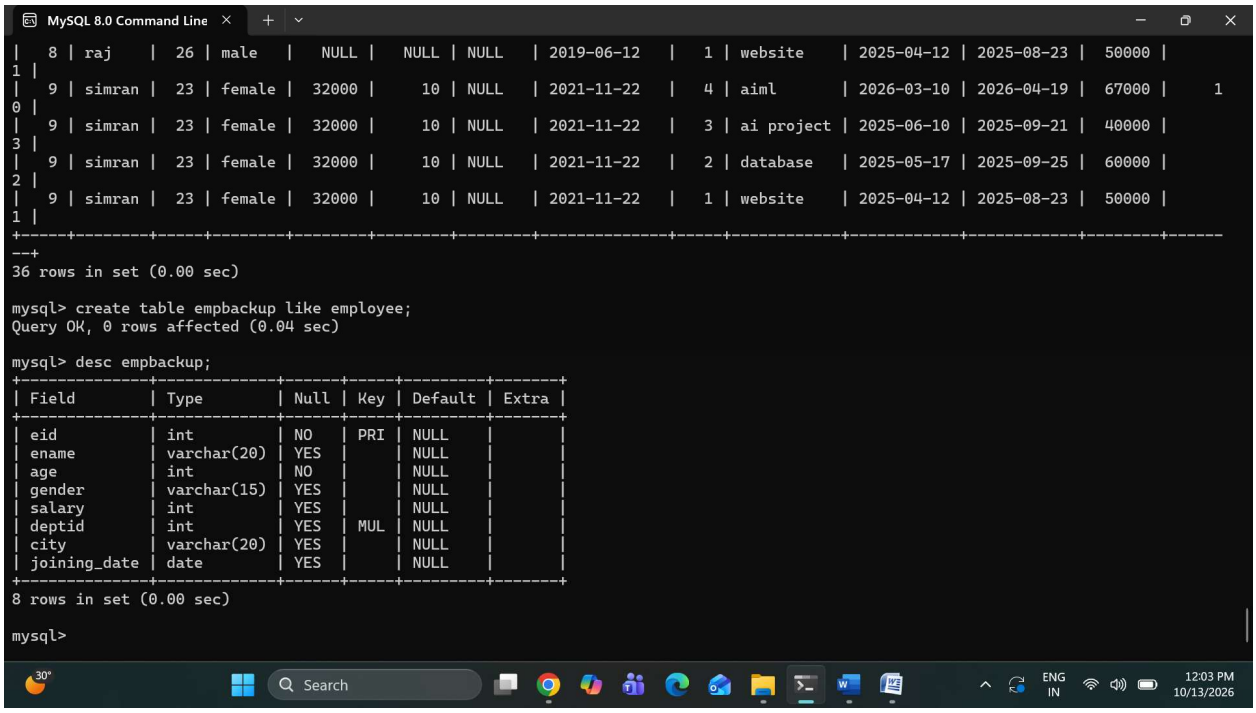

MySQL 8.0 Command Line
mysql> select * from employee cross join project;
| eid | ename | age | gender | salary | deptid | city | joining_date | pid | pname | start_date | end_date | budget | depti
d |
| 1 | payal | 25 | female | 27500 | 10 | Mumbai | 2026-10-01 | 4 | aiml | 2026-03-10 | 2026-04-19 | 67000 | 1
0 |
| 1 | payal | 25 | female | 27500 | 10 | Mumbai | 2026-10-01 | 3 | ai project | 2025-06-10 | 2025-09-21 | 40000 |
3 |
| 1 | payal | 25 | female | 27500 | 10 | Mumbai | 2026-10-01 | 2 | database | 2025-05-17 | 2025-09-25 | 60000 |
2 |
| 1 | payal | 25 | female | 27500 | 10 | Mumbai | 2026-10-01 | 1 | website | 2025-04-12 | 2025-08-23 | 50000 |
1 |
| 2 | sujal | 23 | male | 22000 | 10 | goa | 2026-10-05 | 4 | aiml | 2026-03-10 | 2026-04-19 | 67000 | 1
0 |
| 2 | sujal | 23 | male | 22000 | 10 | goa | 2026-10-05 | 3 | ai project | 2025-06-10 | 2025-09-21 | 40000 |
3 |
| 2 | sujal | 23 | male | 22000 | 10 | goa | 2026-10-05 | 2 | database | 2025-05-17 | 2025-09-25 | 60000 |
2 |
| 2 | sujal | 23 | male | 22000 | 10 | goa | 2026-10-05 | 1 | website | 2025-04-12 | 2025-08-23 | 50000 |
1 |
| 3 | riya | 28 | female | 50000 | 3 | pune | 2026-10-10 | 4 | aiml | 2026-03-10 | 2026-04-19 | 67000 | 1
0 |
| 3 | riya | 28 | female | 50000 | 3 | pune | 2026-10-10 | 3 | ai project | 2025-06-10 | 2025-09-21 | 40000 |
3 |
| 3 | riya | 28 | female | 50000 | 3 | pune | 2026-10-10 | 2 | database | 2025-05-17 | 2025-09-25 | 60000 |
2 |
| 3 | riya | 28 | female | 50000 | 3 | pune | 2026-10-10 | 1 | website | 2025-04-12 | 2025-08-23 | 50000 |
1 |
| 4 | rohit | 28 | male | NULL | 1 | mumbai | 2026-09-10 | 4 | aiml | 2026-03-10 | 2026-04-19 | 67000 | 1
Search
ENG
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10/13/2026


MySQL 8.0 Command Line
1 |
| 4 | rohit | 28 | male | NULL | 1 | mumbai | 2026-09-10 | 4 | aiml | 2026-03-10 | 2026-04-19 | 67000 | 1
0 |
| 4 | rohit | 28 | male | NULL | 1 | mumbai | 2026-09-10 | 3 | ai project | 2025-06-10 | 2025-09-21 | 40000 |
3 |
| 4 | rohit | 28 | male | NULL | 1 | mumbai | 2026-09-10 | 2 | database | 2025-05-17 | 2025-09-25 | 60000 |
2 |
| 4 | rohit | 28 | male | NULL | 1 | mumbai | 2026-09-10 | 1 | website | 2025-04-12 | 2025-08-23 | 50000 |
1 |
| 5 | ojal | 45 | male | 56000 | 3 | pune | 2026-09-23 | 4 | aiml | 2026-03-10 | 2026-04-19 | 67000 | 1
0 |
| 5 | ojal | 45 | male | 56000 | 3 | pune | 2026-09-23 | 3 | ai project | 2025-06-10 | 2025-09-21 | 40000 |
3 |
| 5 | ojal | 45 | male | 56000 | 3 | pune | 2026-09-23 | 2 | database | 2025-05-17 | 2025-09-25 | 60000 |
2 |
| 5 | ojal | 45 | male | 56000 | 3 | pune | 2026-09-23 | 1 | website | 2025-04-12 | 2025-08-23 | 50000 |
1 |
| 6 | aman | 24 | male | 25000 | NULL | NULL | 2020-01-02 | 4 | aiml | 2026-03-10 | 2026-04-19 | 67000 | 1
0 |
| 6 | aman | 24 | male | 25000 | NULL | NULL | 2020-01-02 | 3 | ai project | 2025-06-10 | 2025-09-21 | 40000 |
3 |
| 6 | aman | 24 | male | 25000 | NULL | NULL | 2020-01-02 | 2 | database | 2025-05-17 | 2025-09-25 | 60000 |
2 |
| 6 | aman | 24 | male | 25000 | NULL | NULL | 2020-01-02 | 1 | website | 2025-04-12 | 2025-08-23 | 50000 |
1 |
| 7 | neha | 25 | female | 30000 | 10 | NULL | 2020-06-12 | 4 | aiml | 2026-03-10 | 2026-04-19 | 67000 | 1
0 |
| 7 | neha | 25 | female | 30000 | 10 | NULL | 2020-06-12 | 3 | ai project | 2025-06-10 | 2025-09-21 | 40000 |
3 |
| 7 | neha | 25 | female | 30000 | 10 | NULL | 2020-06-12 | 2 | database | 2025-05-17 | 2025-09-25 | 60000 |
2 |
| 7 | neha | 25 | female | 30000 | 10 | NULL | 2020-06-12 | 1 | website | 2025-04-12 | 2025-08-23 | 50000 |
1 |
Search
ENG
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10/13/2026

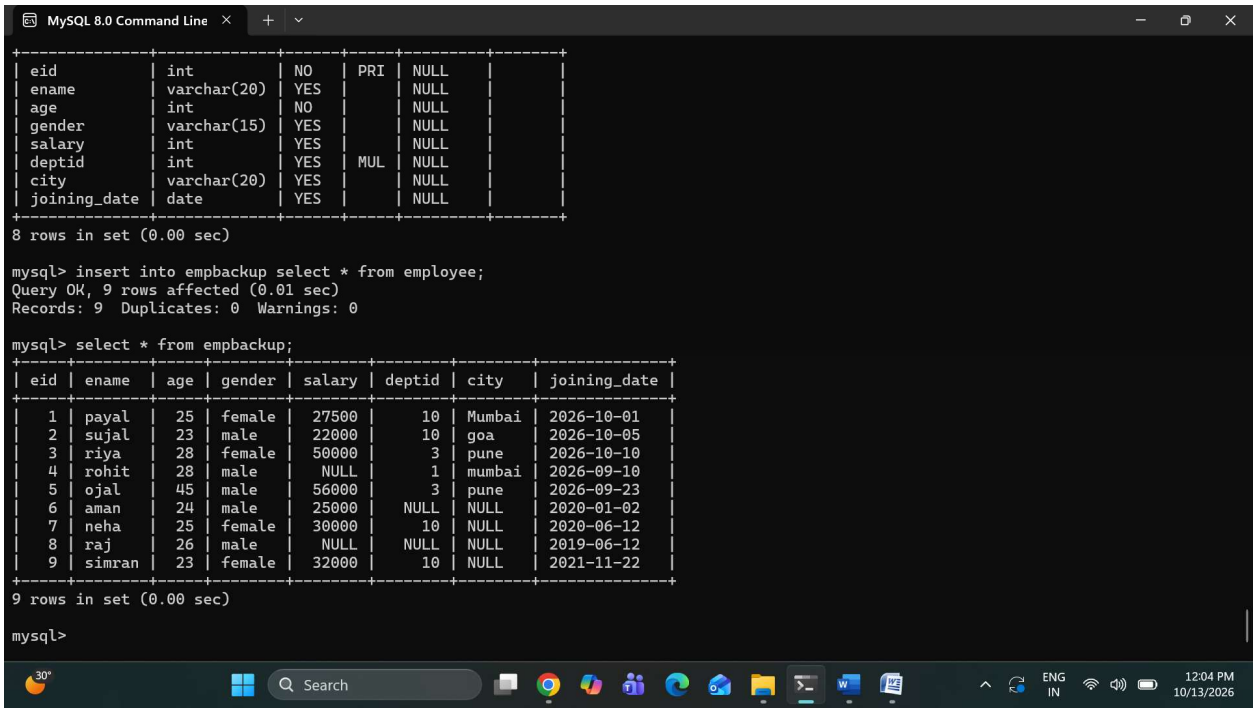

MySQL 8.0 Command Line
1 |
| 6 | aman | 24 | male | 25000 | NULL | NULL | 2020-01-02 | 4 | aiml | 2026-03-10 | 2026-04-19 | 67000 | 1
0 |
| 6 | aman | 24 | male | 25000 | NULL | NULL | 2020-01-02 | 3 | ai project | 2025-06-10 | 2025-09-21 | 40000 |
3 |
| 6 | aman | 24 | male | 25000 | NULL | NULL | 2020-01-02 | 2 | database | 2025-05-17 | 2025-09-25 | 60000 |
2 |
| 6 | aman | 24 | male | 25000 | NULL | NULL | 2020-01-02 | 1 | website | 2025-04-12 | 2025-08-23 | 50000 |
1 |
| 7 | neha | 25 | female | 30000 | 10 | NULL | 2020-06-12 | 4 | aiml | 2026-03-10 | 2026-04-19 | 67000 | 1
0 |
| 7 | neha | 25 | female | 30000 | 10 | NULL | 2020-06-12 | 3 | ai project | 2025-06-10 | 2025-09-21 | 40000 |
3 |
| 7 | neha | 25 | female | 30000 | 10 | NULL | 2020-06-12 | 2 | database | 2025-05-17 | 2025-09-25 | 60000 |
2 |
| 7 | neha | 25 | female | 30000 | 10 | NULL | 2020-06-12 | 1 | website | 2025-04-12 | 2025-08-23 | 50000 |
1 |
| 8 | raj | 26 | male | NULL | NULL | NULL | 2019-06-12 | 4 | aiml | 2026-03-10 | 2026-04-19 | 67000 | 1
0 |
| 8 | raj | 26 | male | NULL | NULL | NULL | 2019-06-12 | 3 | ai project | 2025-06-10 | 2025-09-21 | 40000 |
3 |
| 8 | raj | 26 | male | NULL | NULL | NULL | 2019-06-12 | 2 | database | 2025-05-17 | 2025-09-25 | 60000 |
2 |
| 8 | raj | 26 | male | NULL | NULL | NULL | 2019-06-12 | 1 | website | 2025-04-12 | 2025-08-23 | 50000 |
1 |
| 9 | simran | 23 | female | 32000 | 10 | NULL | 2021-11-22 | 4 | aiml | 2026-03-10 | 2026-04-19 | 67000 | 1
0 |
| 9 | simran | 23 | female | 32000 | 10 | NULL | 2021-11-22 | 3 | ai project | 2025-06-10 | 2025-09-21 | 40000 |
3 |
| 9 | simran | 23 | female | 32000 | 10 | NULL | 2021-11-22 | 2 | database | 2025-05-17 | 2025-09-25 | 60000 |
2 |
| 9 | simran | 23 | female | 32000 | 10 | NULL | 2021-11-22 | 1 | website | 2025-04-12 | 2025-08-23 | 50000 |
1 |
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MySQL 8.0 Command Line
| 6 | aman | 24 | male | 25000 | NULL | NULL | 2020-01-02 | 2 | database | 2025-05-17 | 2025-09-25 | 60000 |
2 |
| 6 | aman | 24 | male | 25000 | NULL | NULL | 2020-01-02 | 1 | website | 2025-04-12 | 2025-08-23 | 50000 |
1 |
| 7 | neha | 25 | female | 30000 | 10 | NULL | 2020-06-12 | 4 | aiml | 2026-03-10 | 2026-04-19 | 67000 | 1
0 |
| 7 | neha | 25 | female | 30000 | 10 | NULL | 2020-06-12 | 3 | ai project | 2025-06-10 | 2025-09-21 | 40000 |
3 |
| 7 | neha | 25 | female | 30000 | 10 | NULL | 2020-06-12 | 2 | database | 2025-05-17 | 2025-09-25 | 60000 |
2 |
| 7 | neha | 25 | female | 30000 | 10 | NULL | 2020-06-12 | 1 | website | 2025-04-12 | 2025-08-23 | 50000 |
1 |
| 8 | raj | 26 | male | NULL | NULL | NULL | 2019-06-12 | 4 | aiml | 2026-03-10 | 2026-04-19 | 67000 | 1
0 |
| 8 | raj | 26 | male | NULL | NULL | NULL | 2019-06-12 | 3 | ai project | 2025-06-10 | 2025-09-21 | 40000 |
3 |
| 8 | raj | 26 | male | NULL | NULL | NULL | 2019-06-12 | 2 | database | 2025-05-17 | 2025-09-25 | 60000 |
2 |
| 8 | raj | 26 | male | NULL | NULL | NULL | 2019-06-12 | 1 | website | 2025-04-12 | 2025-08-23 | 50000 |
1 |
| 9 | simran | 23 | female | 32000 | 10 | NULL | 2021-11-22 | 4 | aiml | 2026-03-10 | 2026-04-19 | 67000 | 1
0 |
| 9 | simran | 23 | female | 32000 | 10 | NULL | 2021-11-22 | 3 | ai project | 2025-06-10 | 2025-09-21 | 40000 |
3 |
| 9 | simran | 23 | female | 32000 | 10 | NULL | 2021-11-22 | 2 | database | 2025-05-17 | 2025-09-25 | 60000 |
2 |
| 9 | simran | 23 | female | 32000 | 10 | NULL | 2021-11-22 | 1 | website | 2025-04-12 | 2025-08-23 | 50000 |
1 |
+-----+--------+-----+--------+--------+--------+--------+-------------+-----+------------+------------+------------+--------+------
--+
36 rows in set (0.00 sec)
mysql>
Search
ENG
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11:56 AM
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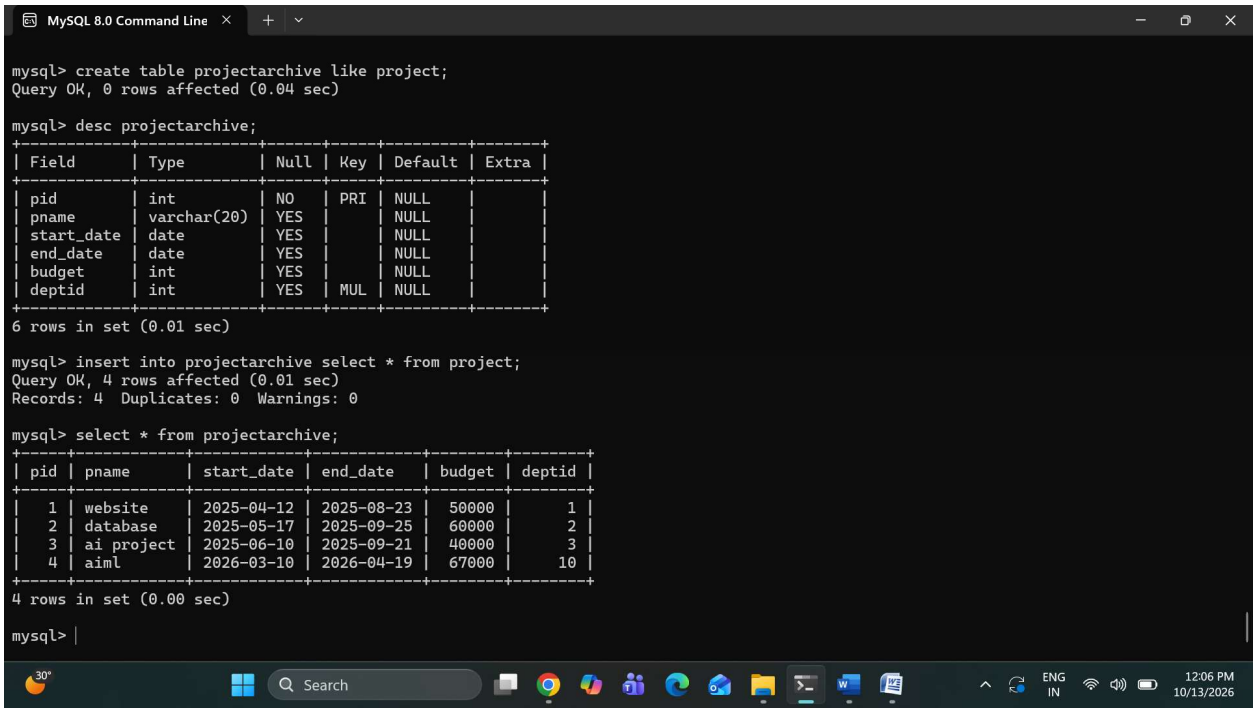
Q7.


MySQL 8.0 Command Line
| 8 | raj | 26 | male | NULL | NULL | NULL | 2019-06-12 | 1 | website | 2025-04-12 | 2025-08-23 | 50000 |
1 |
| 9 | simran | 23 | female | 32000 | 10 | NULL | 2021-11-22 | 4 | aiml | 2026-03-10 | 2026-04-19 | 67000 | 1
0 |
| 9 | simran | 23 | female | 32000 | 10 | NULL | 2021-11-22 | 3 | ai project | 2025-06-10 | 2025-09-21 | 40000 |
3 |
| 9 | simran | 23 | female | 32000 | 10 | NULL | 2021-11-22 | 2 | database | 2025-05-17 | 2025-09-25 | 60000 |
2 |
| 9 | simran | 23 | female | 32000 | 10 | NULL | 2021-11-22 | 1 | website | 2025-04-12 | 2025-08-23 | 50000 |
1 |
--+
36 rows in set (0.00 sec)
mysql> create table empbackup like employee;
Query OK, 0 rows affected (0.04 sec)
mysql> desc empbackup;
Field	Type	Null	Key	Default	Extra
eid	int	NO	PRI	NULL	
ename	varchar(20)	YES		NULL	
age	int	NO		NULL	
gender	varchar(15)	YES		NULL	
salary	int	YES		NULL	
deptid	int	YES	MUL	NULL	
city	varchar(20)	YES		NULL	
joining_date	date	YES		NULL	
8 rows in set (0.00 sec)
mysql>
Search
ENG
IN
12:03 PM
10/13/2026

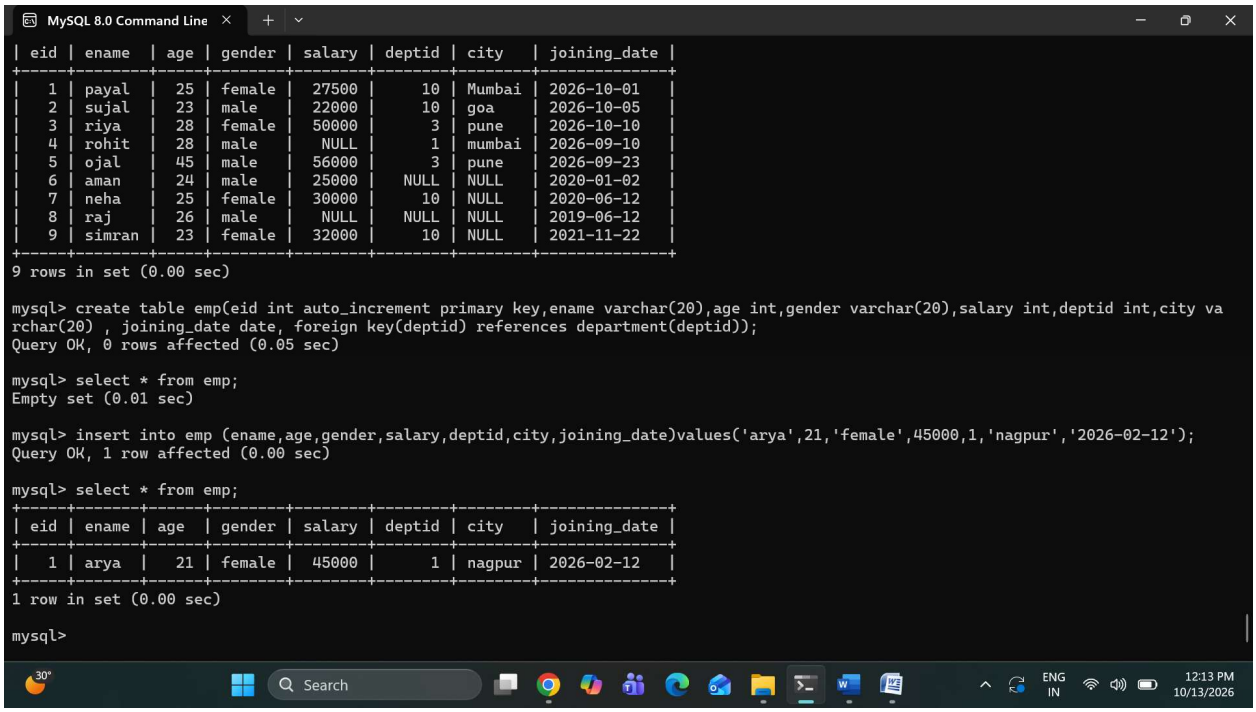
Q8.


MySQL 8.0 Command Line
eid	int	NO	PRI	NULL	
ename	varchar(20)	YES		NULL	
age	int	NO		NULL	
gender	varchar(15)	YES		NULL	
salary	int	YES		NULL	
deptid	int	YES	MUL	NULL	
city	varchar(20)	YES		NULL	
joining_date	date	YES		NULL	
8 rows in set (0.00 sec)
mysql> insert into empbackup select * from employee;
Query OK, 9 rows affected (0.01 sec)
Records: 9 Duplicates: 0 Warnings: 0
mysql> select * from empbackup;
eid	ename	age	gender	salary	deptid	city	joining_date
1	payal	25	female	27500	10	Mumbai	2026-10-01
2	sujal	23	male	22000	10	goa	2026-10-05
3	riya	28	female	50000	3	pune	2026-10-10
4	rohit	28	male	NULL	1	mumbai	2026-09-10
5	ojal	45	male	56000	3	pune	2026-09-23
6	aman	24	male	25000	NULL	NULL	2020-01-02
7	neha	25	female	30000	10	NULL	2020-06-12
8	raj	26	male	NULL	NULL	NULL	2019-06-12
9	simran	23	female	32000	10	NULL	2021-11-22
9 rows in set (0.00 sec)
mysql>
Search
ENG
IN
12:04 PM
10/13/2026

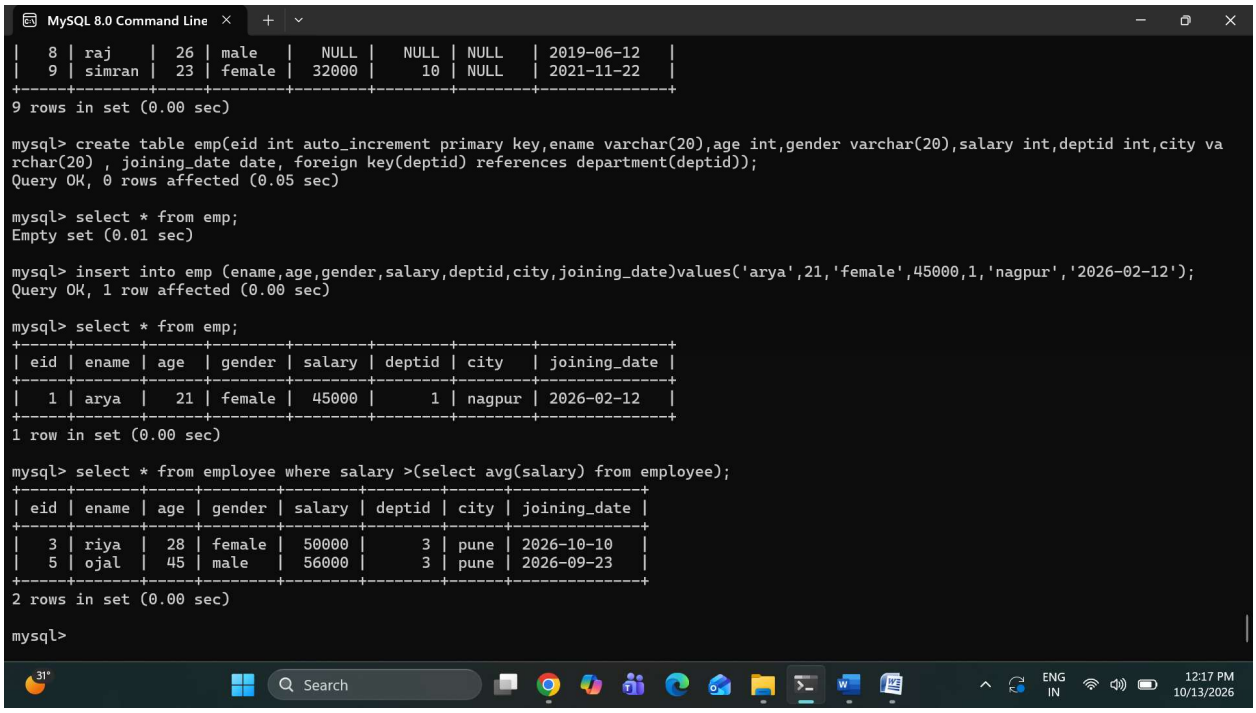
Q9.


MySQL 8.0 Command Line
mysql> create table projectarchive like project;
Query OK, 0 rows affected (0.04 sec)
mysql> desc projectarchive;
+------------+-------------+------+-----+---------+-------+
| Field | Type | Null | Key | Default | Extra |
+------------+-------------+------+-----+---------+-------+
pid	int	NO	PRI	NULL	
pname	varchar(20)	YES		NULL	
start_date	date	YES		NULL	
end_date	date	YES		NULL	
budget	int	YES		NULL	
deptid	int	YES	MUL	NULL	
+------------+-------------+------+-----+---------+-------+
6 rows in set (0.01 sec)
mysql> insert into projectarchive select * from project;
Query OK, 4 rows affected (0.01 sec)
Records: 4 Duplicates: 0 Warnings: 0
mysql> select * from projectarchive;
+-----+------------+------------+------------+--------+--------+
| pid | pname | start_date | end_date | budget | deptid |
+-----+------------+------------+------------+--------+--------+
1	website	2025-04-12	2025-08-23	50000	1
2	database	2025-05-17	2025-09-25	60000	2
3	ai project	2025-06-10	2025-09-21	40000	3
4	aiml	2026-03-10	2026-04-19	67000	10
+-----+------------+------------+------------+--------+--------+
4 rows in set (0.00 sec)
mysql>
Search
ENG
IN
12:06 PM
10/13/2026

Q10.


MySQL 8.0 Command Line
| eid | ename | age | gender | salary | deptid | city | joining_date |
+-----+--------+-----+--------+--------+--------+--------+--------------+
1	payal	25	female	27500	10	Mumbai	2026-10-01
2	sujal	23	male	22000	10	goa	2026-10-05
3	riya	28	female	50000	3	pune	2026-10-10
4	rohit	28	male	NULL	1	mumbai	2026-09-10
5	ojal	45	male	56000	3	pune	2026-09-23
6	aman	24	male	25000	NULL	NULL	2020-01-02
7	neha	25	female	30000	10	NULL	2020-06-12
8	raj	26	male	NULL	NULL	NULL	2019-06-12
9	simran	23	female	32000	10	NULL	2021-11-22
+-----+--------+-----+--------+--------+--------+--------+--------------+
9 rows in set (0.00 sec)
mysql> create table emp(eid int auto_increment primary key,ename varchar(20),age int,gender varchar(20),salary int,deptid int,city va
rchar(20) , joining_date date, foreign key(deptid) references department(deptid));
Query OK, 0 rows affected (0.05 sec)
mysql> select * from emp;
Empty set (0.01 sec)
mysql> insert into emp (ename,age,gender,salary,deptid,city,joining_date)values('arya',21,'female',45000,1,'nagpur','2026-02-12');
Query OK, 1 row affected (0.00 sec)
mysql> select * from emp;
+-----+-------+------+--------+--------+--------+--------+--------------+
| eid | ename | age | gender | salary | deptid | city | joining_date |
+-----+-------+------+--------+--------+--------+--------+--------------+
| 1 | arya | 21 | female | 45000 | 1 | nagpur | 2026-02-12 |
+-----+-------+------+--------+--------+--------+--------+--------------+
1 row in set (0.00 sec)
mysql>
Search
ENG
IN
12:13 PM
10/13/2026

Q11.



```
MySQL 8.0 Command Line
|   8 | raj    |  26 | male   |   NULL |   NULL | NULL   | 2019-06-12   |
|   9 | simran |  23 | female |  32000 |     10 | NULL   | 2021-11-22   |
+-----+--------+-----+--------+--------+--------+--------+--------------+
9 rows in set (0.00 sec)

mysql> create table emp(eid int auto_increment primary key,ename varchar(20),age int,gender varchar(20),salary int,deptid int,city va
rchar(20) , joining_date date, foreign key(deptid) references department(deptid));
Query OK, 0 rows affected (0.05 sec)

mysql> select * from emp;
Empty set (0.01 sec)

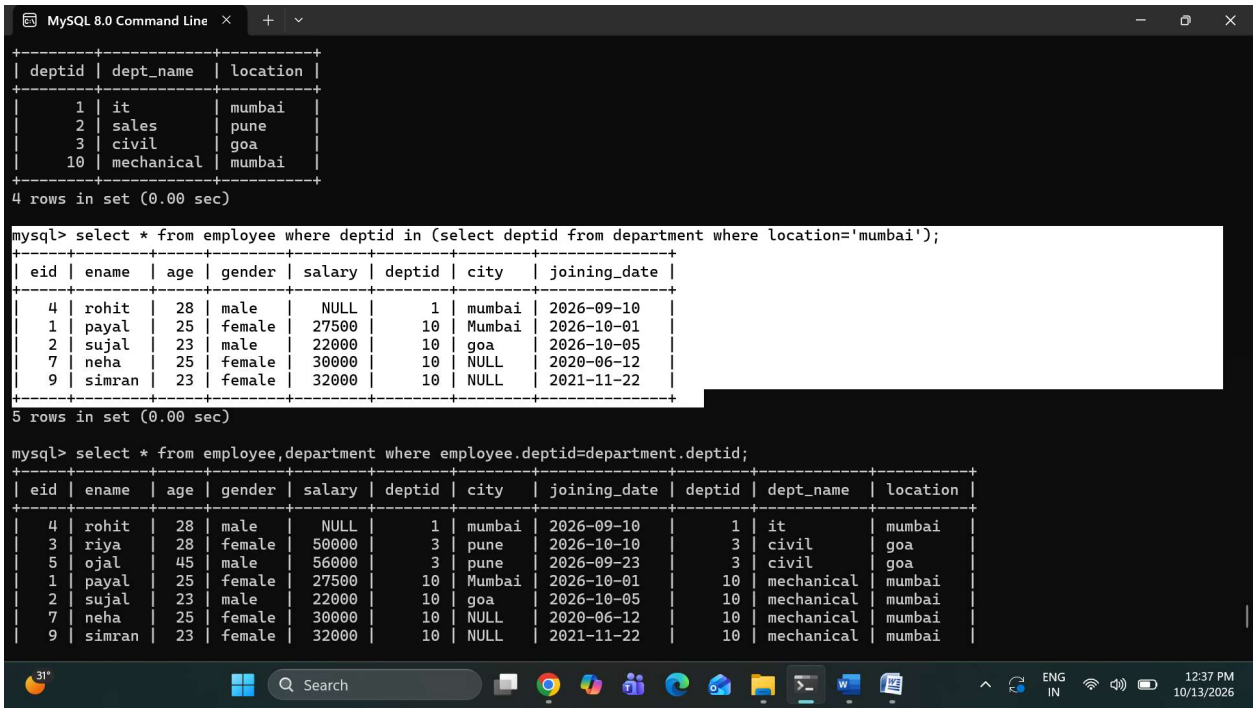
mysql> insert into emp (ename,age,gender,salary,deptid,city,joining_date)values('arya',21,'female',45000,1,'nagpur','2026-02-12');
Query OK, 1 row affected (0.00 sec)

mysql> select * from emp;
+-----+-------+------+--------+--------+--------+--------+--------------+
| eid | ename | age  | gender | salary | deptid | city   | joining_date |
+-----+-------+------+--------+--------+--------+--------+--------------+
|   1 | arya  |   21 | female |  45000 |      1 | nagpur | 2026-02-12   |
+-----+-------+------+--------+--------+--------+--------+--------------+
1 row in set (0.00 sec)

mysql> select * from employee where salary >(select avg(salary) from employee);
+-----+-------+-----+--------+--------+--------+------+--------------+
| eid | ename | age | gender | salary | deptid | city | joining_date |
+-----+-------+-----+--------+--------+--------+------+--------------+
|   3 | riya  |  28 | female |  50000 |      3 | pune | 2026-10-10   |
|   5 | ojal  |  45 | male   |  56000 |      3 | pune | 2026-09-23   |
+-----+-------+-----+--------+--------+--------+------+--------------+
2 rows in set (0.00 sec)

mysql>
```

Q12.

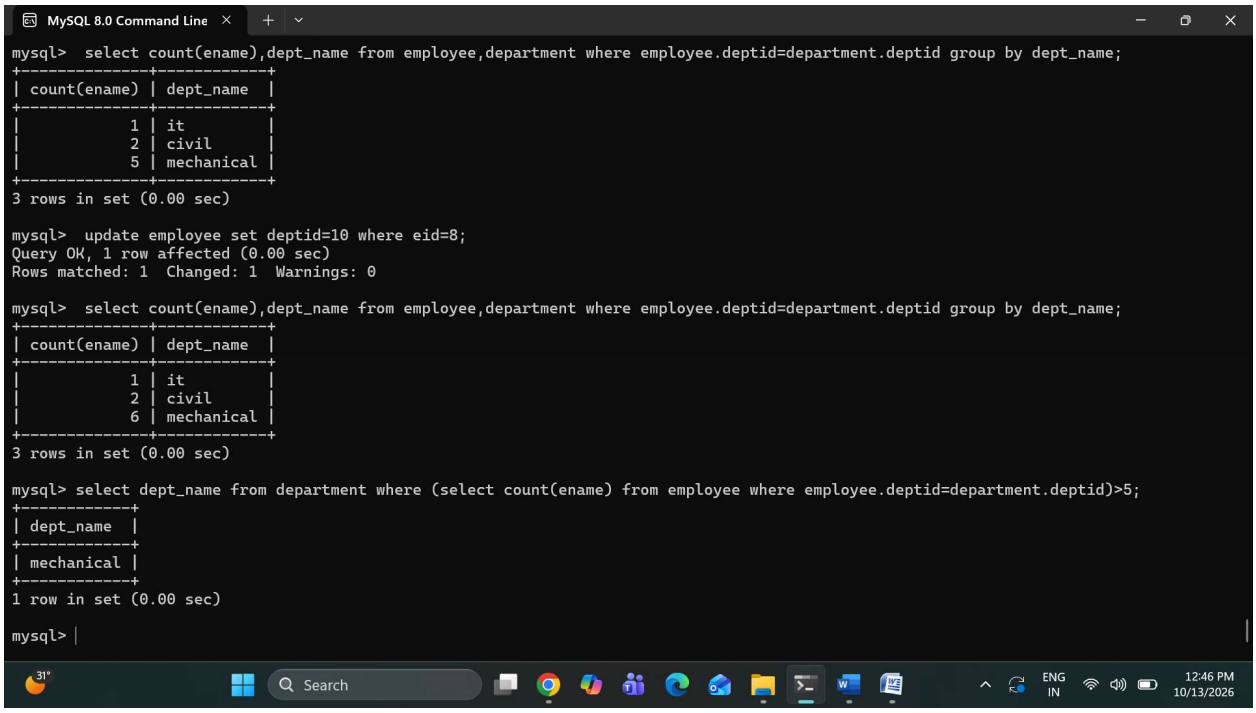


```
MySQL 8.0 Command Line
+--------+------------+----------+
| deptid | dept_name  | location |
+--------+------------+----------+
|      1 | it         | mumbai   |
|      2 | sales      | pune     |
|      3 | civil      | goa      |
|     10 | mechanical | mumbai   |
+--------+------------+----------+
4 rows in set (0.00 sec)

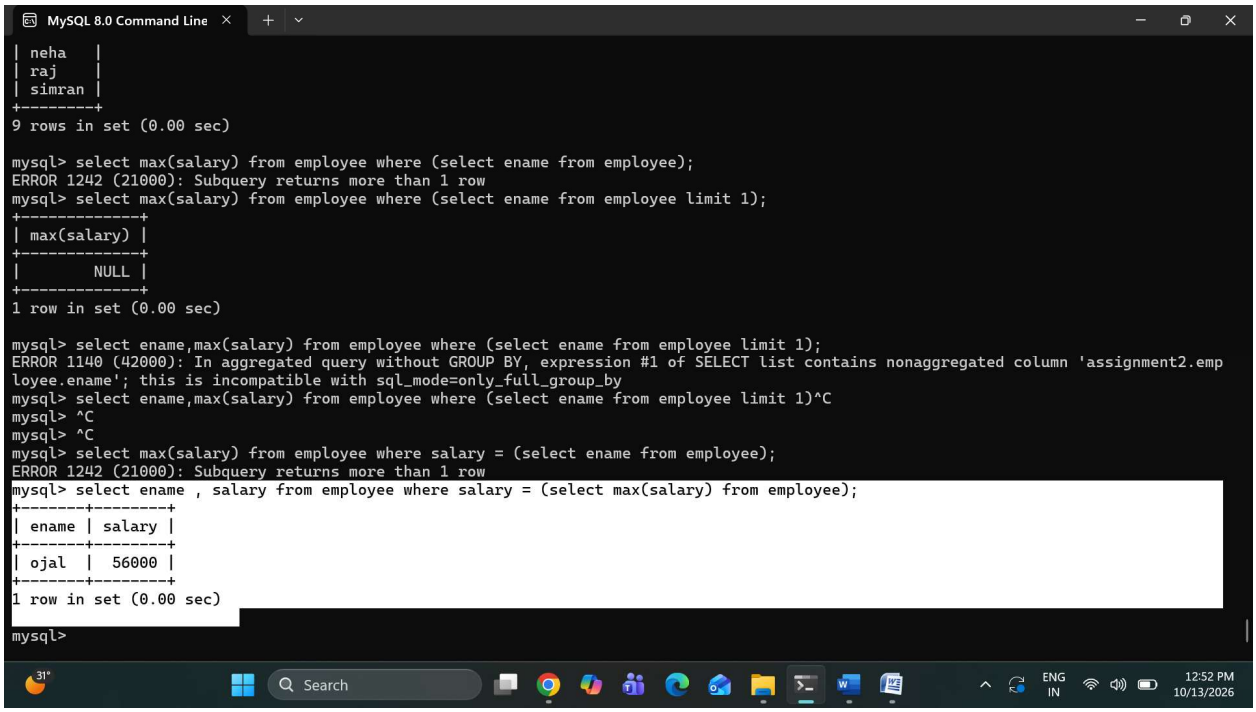
mysql> select * from employee where deptid in (select deptid from department where location='mumbai');
+-----+--------+-----+--------+--------+--------+--------+--------------+
| eid | ename  | age | gender | salary | deptid | city   | joining_date |
+-----+--------+-----+--------+--------+--------+--------+--------------+
|   4 | rohit  |  28 | male   |   NULL |      1 | mumbai | 2026-09-10   |
|   1 | payal  |  25 | female |  27500 |     10 | Mumbai | 2026-10-01   |
|   2 | sujal  |  23 | male   |  22000 |     10 | goa    | 2026-10-05   |
|   7 | neha   |  25 | female |  30000 |     10 | NULL   | 2020-06-12   |
|   9 | simran |  23 | female |  32000 |     10 | NULL   | 2021-11-22   |
+-----+--------+-----+--------+--------+--------+--------+--------------+
5 rows in set (0.00 sec)

mysql> select * from employee,department where employee.deptid=department.deptid;
+-----+--------+-----+--------+--------+--------+--------+--------------+--------+------------+----------+
| eid | ename  | age | gender | salary | deptid | city   | joining_date | deptid | dept_name  | location |
+-----+--------+-----+--------+--------+--------+--------+--------------+--------+------------+----------+
|   4 | rohit  |  28 | male   |   NULL |      1 | mumbai | 2026-09-10   |      1 | it         | mumbai   |
|   3 | riya   |  28 | female |  50000 |      3 | pune   | 2026-10-10   |      3 | civil      | goa      |
|   5 | ojal   |  45 | male   |  56000 |      3 | pune   | 2026-09-23   |      3 | civil      | goa      |
|   1 | payal  |  25 | female |  27500 |     10 | Mumbai | 2026-10-01   |     10 | mechanical | mumbai   |
|   2 | sujal  |  23 | male   |  22000 |     10 | goa    | 2026-10-05   |     10 | mechanical | mumbai   |
|   7 | neha   |  25 | female |  30000 |     10 | NULL   | 2020-06-12   |     10 | mechanical | mumbai   |
|   9 | simran |  23 | female |  32000 |     10 | NULL   | 2021-11-22   |     10 | mechanical | mumbai   |
```

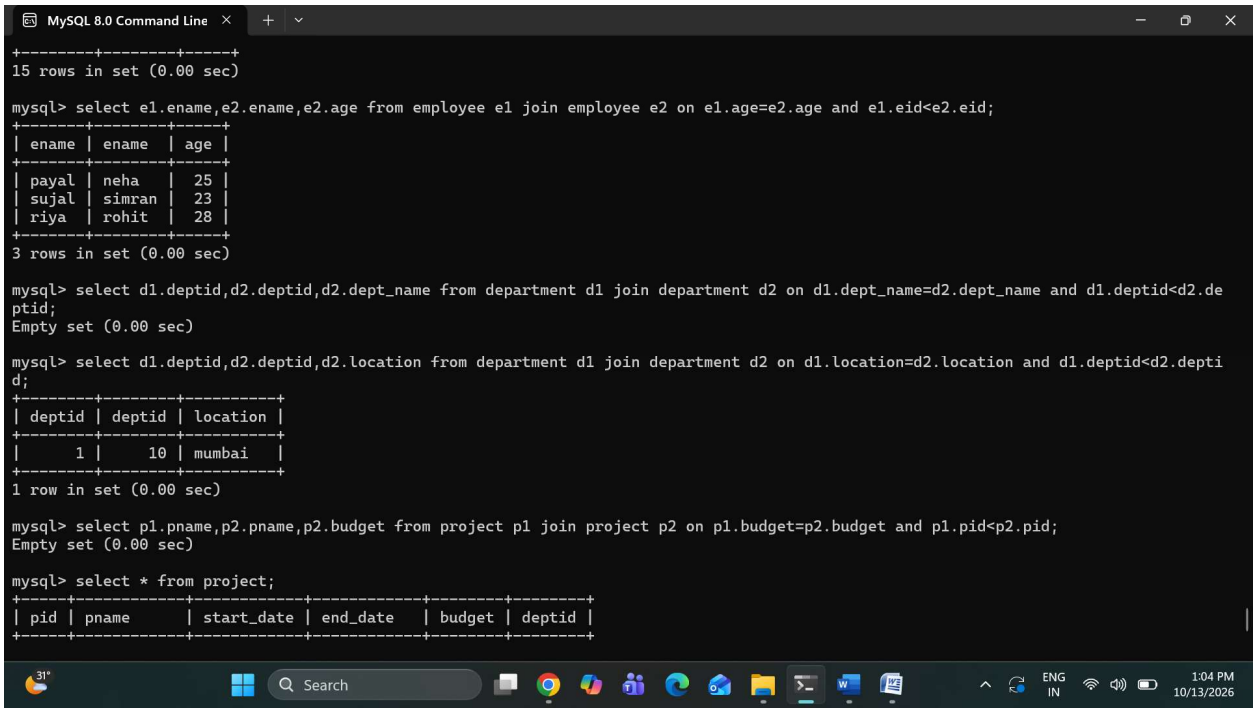
Q13.

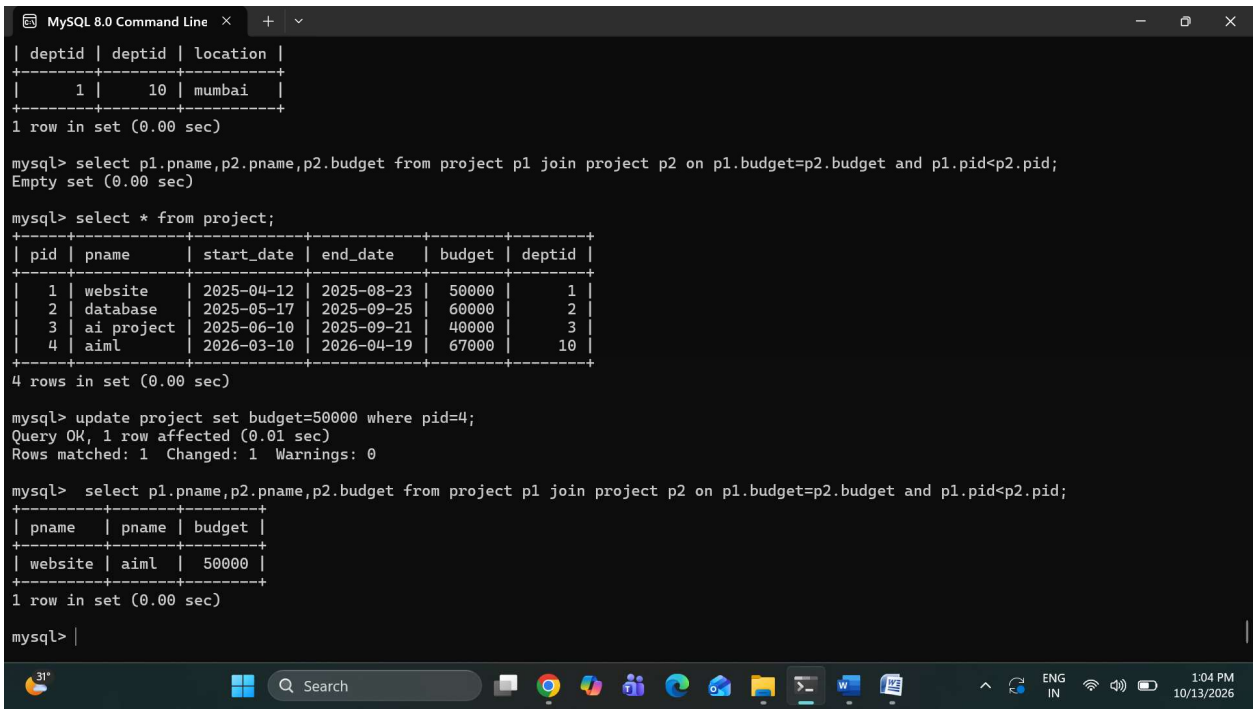

MySQL 8.0 Command Line
mysql> select count(ename),dept_name from employee,department where employee.deptid=department.deptid group by dept_name;
+--------------+------------+
| count(ename) | dept_name |
+--------------+------------+
1	it
2	civil
5	mechanical
+--------------+------------+
3 rows in set (0.00 sec)
mysql> update employee set deptid=10 where eid=8;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0
mysql> select count(ename),dept_name from employee,department where employee.deptid=department.deptid group by dept_name;
+--------------+------------+
| count(ename) | dept_name |
+--------------+------------+
1	it
2	civil
6	mechanical
+--------------+------------+
3 rows in set (0.00 sec)
mysql> select dept_name from department where (select count(ename) from employee where employee.deptid=department.deptid)>5;
+------------+
| dept_name |
+------------+
| mechanical |
+------------+
1 row in set (0.00 sec)
mysql>
Search
ENG
IN
12:46 PM
10/13/2026

Q14.


MySQL 8.0 Command Line
| neha |
| raj |
| simran |
+--------+
9 rows in set (0.00 sec)
mysql> select max(salary) from employee where (select ename from employee);
ERROR 1242 (21000): Subquery returns more than 1 row
mysql> select max(salary) from employee where (select ename from employee limit 1);
+-------------+
| max(salary) |
+-------------+
| NULL |
+-------------+
1 row in set (0.00 sec)
mysql> select ename,max(salary) from employee where (select ename from employee limit 1);
ERROR 1140 (42000): In aggregated query without GROUP BY, expression #1 of SELECT list contains nonaggregated column 'assignment2.emp
loyee.ename'; this is incompatible with sql_mode=only_full_group_by
mysql> select ename,max(salary) from employee where (select ename from employee limit 1)^C
mysql> ^C
mysql> ^C
mysql> select max(salary) from employee where salary = (select ename from employee);
ERROR 1242 (21000): Subquery returns more than 1 row
mysql> select ename , salary from employee where salary = (select max(salary) from employee);
+--------+--------+
| ename | salary |
+--------+--------+
| ojal | 56000 |
+--------+--------+
1 row in set (0.00 sec)
mysql>
Search
ENG
IN
12:52 PM
10/13/2026

Q15.


MySQL 8.0 Command Line
+--------+--------+-----+
15 rows in set (0.00 sec)
mysql> select e1.ename,e2.ename,e2.age from employee e1 join employee e2 on e1.age=e2.age and e1.eid<e2.eid;
+-------+--------+-----+
| ename | ename | age |
+-------+--------+-----+
payal	neha	25
sujal	simran	23
riya	rohit	28
+-------+--------+-----+
3 rows in set (0.00 sec)
mysql> select d1.deptid,d2.deptid,d2.dept_name from department d1 join department d2 on d1.dept_name=d2.dept_name and d1.deptid<d2.de
ptid;
Empty set (0.00 sec)
mysql> select d1.deptid,d2.deptid,d2.location from department d1 join department d2 on d1.location=d2.location and d1.deptid<d2.depti
d;
+--------+--------+----------+
| deptid | deptid | location |
+--------+--------+----------+
| 1 | 10 | mumbai |
+--------+--------+----------+
1 row in set (0.00 sec)
mysql> select p1.pname,p2.pname,p2.budget from project p1 join project p2 on p1.budget=p2.budget and p1.pid<p2.pid;
Empty set (0.00 sec)
mysql> select * from project;
+-----+------------+------------+------------+--------+--------+
| pid | pname | start_date | end_date | budget | deptid |
+-----+------------+------------+------------+--------+--------+
Search
ENG
IN
1:04 PM
10/13/2026


MySQL 8.0 Command Line
| deptid | deptid | location |
+--------+--------+----------+
| 1 | 10 | mumbai |
+--------+--------+----------+
1 row in set (0.00 sec)
mysql> select p1.pname,p2.pname,p2.budget from project p1 join project p2 on p1.budget=p2.budget and p1.pid<p2.pid;
Empty set (0.00 sec)
mysql> select * from project;
+-----+------------+------------+------------+--------+--------+
| pid | pname | start_date | end_date | budget | deptid |
+-----+------------+------------+------------+--------+--------+
1	website	2025-04-12	2025-08-23	50000	1
2	database	2025-05-17	2025-09-25	60000	2
3	ai project	2025-06-10	2025-09-21	40000	3
4	aiml	2026-03-10	2026-04-19	67000	10
+-----+------------+------------+------------+--------+--------+
4 rows in set (0.00 sec)
mysql> update project set budget=50000 where pid=4;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1 Changed: 1 Warnings: 0
mysql> select p1.pname,p2.pname,p2.budget from project p1 join project p2 on p1.budget=p2.budget and p1.pid<p2.pid;
+---------+-------+--------+
| pname | pname | budget |
+---------+-------+--------+
| website | aiml | 50000 |
+---------+-------+--------+
1 row in set (0.00 sec)
mysql>
Search
ENG
IN
1:04 PM
10/13/2026